

ECON 131

PUBLIC ECONOMICS

Midterm 2 — Study Guide

Spring 2026 • Prof. Danny Yagan

Exam: Monday April 6, 5:10–6:10pm • 75 minutes • 150 points

EXAM FORMAT

Section 1: True / False / Uncertain (6 statements × 10 pts = **60 pts**)

Section 2: Empirical Application (**30 pts**)

Section 3: Analytic / Math Problem (**60 pts**)

Rules: No notes, no calculator, no phone. Written on paper. Show all work.

What This Exam Covers

T/F/U (60 pts): Lectures 08–14 — Labor supply evidence, business tax, savings/capital tax, externalities, public goods, local public goods

Empirical (30 pts): Lectures 08–14 + Lecture 08B — Event study / DD regression representations

Analytic (60 pts): Lectures 07–11 — Optimal labor tax, business tax math, savings models, externalities

⚠ KEY DIFFERENCE FROM MIDTERM 1

The professor said Midterm 2 will be harder than Midterm 1. The T/F/U section is now 60 pts (was 50), the analytic section is 60 pts (was 50), and the empirical section is 30 pts (was 50). This means more weight on conceptual understanding and math.

Course Map: What You Need to Learn

The following lectures make up the Midterm 2 scope. Priority indicates how heavily each is likely tested.

#	Lecture	Priority	What You Need to Know
L07	Optimal Labor Tax	HIGH	Optimal top tax rate formula, Laffer curve, EITC design, extensive vs. intensive margin, equity-efficiency trade-off
L08	Labor Supply Evidence	HIGH	Real vs. avoidance responses, EITC empirical effects, taxable income elasticity estimates, bunching
L08B	Event Study / DD Regression	HIGH	DD regression equation $Y = \alpha + \beta T + \gamma \text{Post} + \delta(T \times \text{Post})$, coefficient interpretation
L09	Business Tax	HIGH	Cost-of-capital model, net vs. gross profit tax, traditional vs. new view of dividends, C-corp vs. S-corp, tax avoidance
L10	Taxes & Savings	MEDIUM	Life-cycle savings model, substitution vs. income effect on savings, consumption tax equivalence, wealth inequality, estate tax, bequest motives
L11	Externalities	HIGH	SMC vs. PMC, Coase theorem, corrective taxes vs. permits, DWL from externalities, climate change policy
L12	Public Goods	MEDIUM	Samuelson rule, vertical summation, free rider problem, crowding out, charitable giving experiments
L14	Local Public Goods	MEDIUM	Tiebout model, property tax capitalization, Prop 13, intergovernmental grants, flypaper effect, racial assessment disparities

L07 — Optimal Labor Income Tax (Analytic Focus)

Core Ideas in Plain English

This lecture asks: how much should the government tax people's earnings to pay for transfers to the poor? The answer depends on a trade-off. If you tax high earners more, you can give more to the poor (equity). But high earners may work less in response, shrinking the total economic pie (efficiency). The key number that determines this trade-off is the taxable income elasticity — how much people reduce their reported income when taxes go up.

Without any behavioral response (people work the same no matter what), a utilitarian government would set taxes so everyone ends up with exactly the same consumption. But in the real world, people DO respond to taxes, so 100% redistribution would kill work incentives. The optimal tax rate balances these two forces.

Key Concepts

1. The Budget Set and Tax Definitions

After-tax income: $c = z - T(z)$ where z = pre-tax earnings and $T(z)$ = tax/transfer function

Transfer to zero-earners: $-T(0)$ — the amount someone with \$0 earnings receives from the government

Marginal tax rate: $T'(z)$ — how much extra tax you pay for earning \$1 more

Participation tax rate: $\tau_p = [T(z) - T(0)] / z$ — the fraction of earnings lost when moving from zero earnings to earnings z

2. Intensive vs. Extensive Margin

Intensive margin: how many hours you work (or how much income you report), given that you already have a job

Extensive margin: whether you work at all — the decision to participate in the labor force or stay out

Why this matters: The EITC is good policy when the **extensive margin response is large** (it pulls people into the workforce). It is LESS desirable when the intensive margin response is large (the phase-out creates high implicit MTRs for people already working).

MIDTERM 1 TRAP — YOU GOT THIS WRONG LAST TIME

On Midterm 1 Q1(e), the statement was: 'An EITC is desirable when the intensive margin response is larger than the extensive margin response.' The correct answer is FALSE. EITC works best when extensive > intensive. You answered correctly (FALSE) but your reasoning was unclear. Make sure you can explain WHY clearly: the EITC subsidizes entering the workforce (extensive), but its phase-out penalizes earning more (intensive).

3. Optimal Top Tax Rate

The government wants to maximize tax revenue from top earners (because their marginal utility of consumption is very small). Two forces operate when the top rate goes up:

Mechanical effect: Government collects more revenue from existing earnings → good

Behavioral effect: Top earners reduce reported income → bad (lost revenue)

4. Utilitarian Result with No Behavioral Response

If nobody changes their behavior in response to taxes, a utilitarian government (maximizing sum of utilities) with diminishing marginal utility will equalize everyone's consumption. This means 100% marginal tax rate and full redistribution.

MIDTERM 1 TRAP — YOU GOT THIS WRONG

On Midterm 1 Q1(d): 'If taxes and transfers have absolutely no behavioral effects, a utilitarian government sets taxes so everyone consumes the same amount.' The answer is TRUE. You said FALSE. The logic: with no behavioral response, there is zero efficiency cost to redistribution, so diminishing marginal utility means the government maximizes total welfare by equalizing consumption.

Formulas You Must Know

Formula	Meaning
$c = (1-t)wl + T$	Household budget: consumption = after-tax wage $\times$ labor + lump-sum transfer
$l^* = [(1-t)w]^\eta$	Optimal labor supply from standard utility model. Higher after-tax wage $\rightarrow$ more work.
$c^* = [(1-t)w]^{(1+\eta)} + T$	Optimal consumption: plug l^* into budget constraint
$TIE = \eta$	Taxable income elasticity equals the curvature parameter. Empirically 0.1–0.5.
$R(\tau) = \tau \cdot Z(1-\tau)$	Tax revenue as function of tax rate (Laffer curve). Inverted U-shape.
$\tau^* = 1/(1+e)$	Revenue-maximizing rate. e = elasticity of average income. Higher $e \rightarrow$ lower optimal rate.
$\tau^* = 1/(1+ae)$	Optimal top rate. $a = \bar{z}/(\bar{z}-z^*) \approx 1.5$ (thickness of top tail). $e \approx 0.25$ –0.5.
$\tau = (1-\bar{a})/(1-\bar{a}+e)$	Optimal linear tax. $\bar{a}$ = social welfare weight on earners. Close to Laffer rate when inequality high.

L08 — Labor Supply Evidence (T/F/U + Empirical Focus)

Core Ideas in Plain English

This lecture looks at the EVIDENCE for how people actually respond to taxes. Theory says taxes could increase or decrease labor supply (substitution vs. income effects). The empirical question is: which effect wins? And how big is the response?

The key finding: labor supply responses are generally SMALL on the intensive margin (people don't change hours much) but can be LARGE on the extensive margin (whether to work at all, especially for single mothers responding to EITC). Also, much of the "taxable income response" comes from avoidance and evasion, not actual changes in real work.

Key Concepts

Real vs. Avoidance Responses

Real response: People actually work fewer hours or produce less output when taxes go up

Avoidance response: People shift income across time periods, across tax bases (wage → profit), or across jurisdictions to reduce taxes without changing real work

Evasion response: Illegal underreporting of income (especially cash businesses)

Why this matters for policy: If most of the taxable income elasticity comes from avoidance/evasion rather than real labor supply changes, the government can fix this by **broadening the tax base and better enforcement** rather than lowering rates.

Key Empirical Findings

EITC and single mothers: Large extensive margin (participation) response. When EITC expanded in the 1990s, single mothers entered the workforce in large numbers. But Kleven (2018) cautions that social norms and economic boom also contributed.

Bunching at EITC kinks: Saez (2010) found that taxpayers bunch (cluster their reported income) at the first EITC kink point, but only self-employed workers do this — wage workers cannot easily manipulate reported income.

High earner migration: Kleven et al. (2013): Denmark gave a tax break to high-earning foreigners. Sharp increase in their migration to Denmark. Shows top earners DO respond to taxes through location choice.

Executive timing: Goolsbee (2000): When the 1993 tax increase was announced, executives exercised stock options BEFORE the new rate kicked in. Large short-run response, small long-run response → timing, not real behavior.

L08B — DD Regression Representation (Empirical Focus)

Core Ideas in Plain English

You already know difference-in-differences (DD) from Midterm 1 — you compare the change in the treatment group to the change in the control group. This lecture teaches you how to express DD as a regression equation, which is how economists actually estimate it with data.

The DD Regression Equation

$$Y_{it} = \alpha + \beta \cdot \text{Treated}_i + \gamma \cdot \text{Post}_t + \delta \cdot (\text{Treated}_i \times \text{Post}_t) + \epsilon_{it}$$

What Each Coefficient Means

Coefficient	Interpretation
α	Average outcome for the CONTROL group in the PRE-treatment period. This is your baseline. In the 2x2 table: $\alpha = \bar{Y}(\text{Control}, \text{Before})$
β	Difference between treatment and control groups BEFORE the policy. If parallel pre-trends hold, this is a fixed level difference that existed before the policy. $\beta = \bar{Y}(\text{Treated}, \text{Before}) - \bar{Y}(\text{Control}, \text{Before})$
γ	The change over time in the CONTROL group. This captures time trends that affected everyone — things that would have happened even without the policy. $\gamma = \bar{Y}(\text{Control}, \text{After}) - \bar{Y}(\text{Control}, \text{Before})$
δ (the DD estimate)	★ THIS IS THE CAUSAL EFFECT. The extra change in the treatment group beyond what the control group experienced. $\delta = [\bar{Y}(T, \text{After}) - \bar{Y}(T, \text{Before})] - [\bar{Y}(C, \text{After}) - \bar{Y}(C, \text{Before})]$

How the Regression Maps to the 2x2 Table

Think of four group means:

	Before (Post=0)	After (Post=1)
Control (Treated=0)	α	$\alpha + \gamma$
Treatment (Treated=1)	$\alpha + \beta$	$\alpha + \beta + \gamma + \delta$

The DD estimate δ isolates the treatment effect by removing both the pre-existing group difference (β) and the common time trend (γ).

Example: NOx Budget Program

Policy: 2003 regulation on summer pollution from power plants in some states.

Treated: Summer (when regulation applies). **Control:** Winter (regulation off).

DD regression: Emissions = $\alpha + \beta \cdot \text{Summer} + \gamma \cdot \text{Post2003} + \delta \cdot (\text{Summer} \times \text{Post2003})$

Result: $\delta = -38$ units. Summer emissions fell by 38 more than winter. Causal effect of regulation.

Example: Denmark High-Earner Tax Break (KLS 2013)

Policy: 1991 Denmark introduced low tax rate for high-earning foreign workers.

Treated: Foreign workers in top bracket (eligible for break). **Control:** Foreign workers just below threshold.

Result: $\delta = +45.89$ additional workers in treated bracket. Tax break attracted high earners.

EXAM TIP: WHAT YAGAN WILL LIKELY TEST

The empirical section (30 pts) will probably give you a policy scenario and ask you to: (1) identify treatment and control groups, (2) write the DD regression, (3) interpret δ , and (4) discuss whether parallel pre-trends are satisfied. Practice mapping real-world policies into the $Y = \alpha + \beta T + \gamma \text{Post} + \delta(T \times \text{Post})$ framework.

L09 — Business Taxation (Analytic + T/F/U Focus)

Core Ideas in Plain English

Businesses earn profits and the government taxes them. The key question is: does taxing business income reduce investment? The answer depends on whether the government lets businesses deduct their costs.

If businesses can deduct ALL costs (including the cost of buying machines and buildings), then the business income tax is really just a tax on pure profits — and taxing pure profits does NOT distort investment. But in reality, capital costs are only partially deductible (through depreciation schedules spread over many years), so the tax DOES reduce investment.

Key Concepts

1. Cost-of-Capital Model (Hall-Jorgenson 1967)

A firm chooses capital K to maximize profits. The first-order condition tells us the optimal investment level:

No tax: $F'(K) = r$ — invest until marginal product of capital equals cost of capital

Tax on gross profits: $F'(K) = r / (1-\tau)$ — tax raises the required return → **reduces investment**

Tax on net profits (full cost deduction): $F'(K) = r$ — tax does NOT distort investment (Robertson 1927)

Plain English: If the government taxes your revenue minus ALL your costs, it is just taking a share of your pure profit. Since the tax does not change the comparison between investing and not investing, you invest the same amount.

2. Traditional vs. New View of Dividend Taxation

Both views agree that the corporate income tax reduces investment. They disagree about whether the DIVIDEND tax on top of that also reduces investment.

	Traditional View	New View
Firm type	New/startup firm raising outside capital	Mature firm reinvesting existing profits
Dividend tax effect	Reduces investment (raises cost of capital)	Does NOT affect investment
Logic	Outside investor demands higher return to cover dividend tax	Opportunity cost of investing (paying dividends) and return on investing are BOTH reduced by same factor → cancels out
FOC	$F'(K) = r / [(1-\tau_{DIV})(1-\tau_{INC})]$	$F'(K) = r / (1-\tau_{INC})$

3. Yagan (2015): Testing Traditional vs. New View

The 2003 US tax cut reduced the dividend tax from 38.6% to 15%. Yagan used DD comparing **C-corporations** (affected by dividend tax) vs. **S-corporations** (not affected, pass-through). Result: **investment elasticity = 0.00**. The new view wins: dividend tax cuts don't increase investment.

4. C-Corp vs. S-Corp (Pass-Through) Taxation

C-corporation: Pays corporate income tax (21%) on profits, THEN shareholders pay dividend/capital gains tax when profits are distributed. Double taxation.

S-corporation / Partnership: Profits "pass through" directly to owners and are taxed only at the individual level. No corporate-level tax.

Tax arbitrage problem: Business owners classify labor income as "pass-through profits" to avoid payroll taxes (Medicare 3.8% surtax). This is a major source of tax avoidance in the US.

5. Multinational Tax Avoidance

Transfer pricing: Multinational firms shift profits to low-tax countries (Ireland, etc.) by having subsidiaries buy/sell to each other at manipulated prices. Result: 20% of US corporate profits end up in tax havens.

Policy fix — Sales apportionment: Allocate global profits based on where SALES occur (customers can't move to Ireland). Already used by US states.

Formulas You Must Know

Formula	Meaning
Profit = Rev - Labor - Inputs - Depreciation - Interest	What gets taxed by the business income tax
No tax: $F'(K) = r$	Invest until marginal product = cost of capital
Gross profit tax: $F'(K) = r/(1-\tau)$	Tax on revenue minus labor only → discourages investment
Net profit tax: $F'(K) = r$	Full capital deduction → no investment distortion
Traditional: $F'(K) = r/[(1-\tau_D)(1-\tau_C)]$	Both corporate and dividend tax reduce investment (new firms)
New view: $F'(K) = r/(1-\tau_C)$	Only corporate tax matters, dividend tax is on past profits (mature firms)

L10 — Taxes on Capital and Savings (T/F/U + Analytic)

Core Ideas in Plain English

Should the government tax the income people earn from their savings (interest, dividends, capital gains)? There is a fundamental tension: capital income is very unequally distributed (rich people own most of the wealth), so equity says tax it. But taxing savings discourages capital accumulation, which hurts economic growth and ultimately workers' wages.

Key Concepts

1. Life-Cycle Savings Model (Two-Period)

Period 1: You work, earn wage $w \cdot l$, consume c_1 , save $s = w \cdot l - c_1$

Period 2: You retire, consume $c_2 = (1+r) \cdot s$ from your savings

Intertemporal budget constraint: $c_1 + c_2/(1+r) = w \cdot l$

With capital income tax τ_K : $c_1 + c_2/[1+r(1-\tau_K)] = w \cdot l$

2. Effect of Capital Tax on Savings: AMBIGUOUS

Substitution effect: Tax lowers the return to saving $\rightarrow$ future consumption is relatively more expensive $\rightarrow$ save LESS

Income effect: Tax makes you poorer overall $\rightarrow$ consume less in BOTH periods $\rightarrow$ save MORE

Net effect: Theoretically ambiguous. c_2 definitely falls. c_1 and savings could go either way.

KEY EQUIVALENCE: CONSUMPTION TAX = LABOR INCOME TAX

A consumption tax and a labor income tax are EQUIVALENT in this model — both reduce the budget constraint uniformly across time periods. A capital income tax is DIFFERENT because it specifically penalizes future consumption relative to present consumption (like a "double tax" on future spending).

3. Why Tax Capital at All? (Limits of Atkinson-Stiglitz)

The baseline result (Atkinson-Stiglitz) says: if you have an optimal income tax, you don't need a separate capital tax. But this breaks down when:

(a) Different rates of return: Richer people earn higher returns on their investments (better financial advice, scale economies). Capital income correlates with underlying ability.

(b) Income shifting: Without capital tax, business owners reclassify labor income as capital income to pay lower rates.

(c) Inheritances: Large share of wealth comes from inheritance, not own savings. Taxing inherited wealth has different efficiency implications.

4. Wealth Inequality Facts

Top 1% owns ~40% of total US wealth. Top 0.1% owns ~20%.

Bottom 90% share fell from ~35% (1980) to ~20% (2012).

Wealth-to-income ratio: Total household wealth $\approx 6 \times$ annual national income.

5. Estate and Inheritance Taxation

US estate tax: 40% rate on estates above \$13 million. Affects less than 0.1% of deaths.

Four bequest motives and their policy implications:

(a) Accidental bequests: People die with wealth they planned to consume. Tax them heavily — no distortion.

(b) Altruistic bequests: Parents care about children's welfare. Tax creates distortion, but still desirable for redistribution.

(c) Strategic bequests: Parents use inheritance threats to get children to visit/help. Tax inheritances as labor income to recipients.

(d) Social pressure bequests: Parents feel obligated to leave bequests. Estate tax actually helps parents (gives them an excuse to keep money).

Formulas You Must Know

Formula	Meaning
$c_1 + c_2/(1+r) = w \cdot 1$	Intertemporal budget constraint (no tax)
$c_1 + c_2/[1+r(1-\tau_K)] = w \cdot 1$	Budget with capital income tax. Only r is affected.
$W_t = W_{t-1} + r \cdot W_{t-1} + E_t - C_t + I_t$	Wealth accumulation: past wealth + returns + earnings – consumption + inheritance
$\beta = s/g$	Long-run wealth-to-income ratio (Piketty). s = savings rate, g = growth rate.

L11 — Externalities (Analytic + T/F/U Focus)

Core Ideas in Plain English

An externality is when your action affects someone else but you don't pay for (or get compensated for) that effect. A factory polluting a river imposes costs on swimmers, but the factory doesn't pay for the pollution. Because the factory ignores these costs, it produces too much pollution. This is a market failure — the free market gets the wrong answer.

Key Concepts

1. The Core Framework

Private Marginal Cost (PMC): The cost the producer actually pays for one more unit

Marginal Damage (MD): The cost imposed on others (pollution damage, health costs, etc.)

Social Marginal Cost (SMC): $SMC = PMC + MD$

Market outcome: Producers set $PMB = PMC$ (ignore MD) → **overproduction**

Social optimum: Set $PMB = SMC = PMC + MD$ → lower quantity

Deadweight loss: Triangle between demand and SMC curves, from Q_{optimal} to Q_{market}

2. General Principle

Type of Externality	Example	Market Result
Negative production	Factory pollution	Overproduction
Negative consumption	SUV safety risk to others	Overconsumption
Positive production	Beehives pollinating farms	Underproduction
Positive consumption	Beautiful garden for neighbors	Underconsumption

3. Coase Theorem

Part I: If property rights are well-defined and bargaining is costless, private negotiations lead to the socially optimal quantity regardless of who gets the property rights.

Part II: The efficient quantity is the same no matter who is assigned the property rights (polluter or victim).

Limitations: Fails with (a) many affected parties (global warming affects billions), (b) holdout problem (everyone has veto power), (c) high transaction costs.

4. Government Solutions

Corrective tax (Pigouvian tax): Set tax = marginal damage. Producers then face $PMC + \text{tax} = SMC$. Automatically achieves optimal quantity.

Tradable permits (cap-and-trade): Government sets total quantity allowed, issues permits, firms trade them. Price of permit = marginal damage.

5. Taxes vs. Permits Under Uncertainty

Flat MD curve (e.g., CO₂ / global warming): Use TAX. Damage per unit is roughly constant, so locking in the price is fine. If abatement costs are unexpectedly high, firms pollute a bit more and pay more tax — small welfare loss.

Steep MD curve (e.g., nuclear waste): Use PERMITS. Damage rises sharply with pollution, so you MUST lock in the quantity. If costs are high under a tax, firms might pollute way too much.

6. Key Empirical Study: Chay-Greenstone (2003)

1970 Clean Air Act forced counties above pollution threshold to reduce emissions. DD: counties above threshold (treatment) vs. below (control). Result: regulation reduced infant mortality by ~500 per 100K live births. Strong health benefit.

7. Climate Change

Four challenges: (1) Global — one country's emissions affect everyone. (2) Irreversible — CO₂ lasts centuries. (3) Long-term — costs appear decades later. (4) Uncertain — damage estimates vary wildly.

Discount rate problem: If $r = 4\%$, damages 100 years from now have near-zero present value. Standard economics says climate action has low priority. But low r (caring about future generations) implies aggressive action.

Biggest success so far: Rich-country subsidies for solar/wind R&D drove costs down. China and India now adopt renewables voluntarily because they're cheaper, not because of international agreements.

Formulas You Must Know

Formula	Meaning
$SMC = PMC + MD$	Social marginal cost = private cost + external damage
$DWL = \frac{1}{2} \times MD \times (Q_{mkt} - Q_{opt})$	Deadweight loss triangle from overproduction
Corrective tax $t^* = MD$	Set tax equal to marginal damage per unit
$SMB = PMB - MD$ (consumption ext.)	For negative consumption externality: social benefit is lower than private

L12 — Public Goods (T/F/U Focus)

Core Ideas in Plain English

A public good is something where (1) one person's use doesn't reduce others' ability to use it (non-rival), and (2) you can't stop people from using it (non-excludable). Classic example: national defense. The problem is that nobody wants to pay for it voluntarily — everyone hopes others will pay (free riding). So the private market under-provides public goods, and the government must step in.

Key Concepts

1. Public Good Classification

	Rival	Non-Rival
Excludable	Private good (ice cream)	Impure public good (cable TV)
Non-Excludable	Impure (crowded sidewalk)	Pure public good (defense)

2. Samuelson Rule (Optimal Public Good Provision)

For private goods: each person's MRS = price (demand curves add HORIZONTALLY).

For public goods: everyone consumes the same quantity, so we add up everyone's willingness to pay:

$$\sum \text{MRS}_i = \text{MC} \quad (\text{sum of marginal rates of substitution} = \text{marginal cost})$$

Plain English: Add up how much each person values one more unit of the public good. If the total is greater than the cost, produce more.

Demand curves add VERTICALLY for public goods (not horizontally like private goods).

3. Free Rider Problem

Each person decides based on THEIR OWN benefit vs. cost, ignoring the benefit to others. Since public goods benefit everyone, each person under-contributes. Result: private market provides too little of the public good.

Lab evidence (Marwell-Ames 1981): People contribute ~50% initially but contributions fall with repetition as free-riders are identified.

4. Crowding Out

When the government provides more of a public good, private contributions decrease. In theory, \$1 of government provision crowds out \$1 of private provision (one-for-one). In practice, crowding out is partial.

Hungerman (2005): When Clinton cut welfare in 1996, churches in affected areas increased spending by 40 cents per \$1 cut. Partial crowding in.

Andreoni-Payne (2003): \$1 more government grant = 56 cents less private contributions. 70% of this is because charities reduce fundraising effort.

5. Social Pressure and Charitable Giving

DellaVigna-List-Malmendier (2012): Door-to-door fundraising experiment. Giving people advance notice of the fundraiser (so they can avoid opening the door) reduced giving by 28-42%. Implies much giving is driven by social pressure, not altruism.

Falk (2007): Including a small gift (postcard) in a charity letter increased donations from 12% to 14%. A large gift increased to 21%. Reciprocity matters.

L14 — Local Public Goods & Fiscal Federalism (T/F/U Focus)

Core Ideas in Plain English

Should public goods be provided by the federal government or by local cities/towns? The Tiebout model says local provision can be efficient because people "vote with their feet" — they move to the town that offers the tax/services package they like best. This creates competition among towns, just like competition among firms in a market.

Key Concepts

1. Tiebout Model (1956)

Core mechanism: People choose where to live based on local taxes and public goods. Towns compete for residents. This "shopping" for communities can lead to efficient provision of local public goods.

Tiebout Theorem Part I (Sorting): In equilibrium, similar people cluster together. Families with kids live in the town with good schools; elderly people live in the low-tax town.

Tiebout Theorem Part II (Efficiency): Each town provides the public good level its residents actually want. Efficient outcome.

2. Five Problems with Tiebout

(1) **Moving costs:** Moving is expensive; people can't freely relocate.

(2) **Information:** People don't know the exact tax/service bundle in every town.

(3) **Limited variety:** Not enough towns to match every preference.

(4) **Spillovers:** Public goods spill across town borders (e.g., parks, police).

(5) **Not poll taxes:** Real local taxes are property taxes, not equal per-person taxes. Creates distortions.

3. Property Tax Capitalization

If a town has low taxes and good schools, house prices go UP to reflect the benefit. If a town has high taxes and bad services, house prices go DOWN.

Math: A \$1/year perpetual tax cut is capitalized as $\$1 + \$1/(1+r) + \dots = (1+r)/r$ in house value.

Cellini-Ferreira-Rothstein (2010): School bonds that just passed (vs. just failed) increased house prices by 3–9%. Shows schools were previously under-invested.

4. Prop 13 and School Finance Equalization

Serrano v. Priest (1976): California Supreme Court ruled school funding disparities unconstitutional. Required redistribution across districts. Perverse result: rich districts had no incentive to raise taxes (money goes to state pool). School funding fell.

Proposition 13 (1978): Strict limits on property taxes. Assessed value can only grow 2%/year regardless of market value. Creates massive tax differences between long-time and new homeowners. Lock-in effect: people don't move to avoid reassessment.

5. Why People Hate the Property Tax (Wong 2024)

Property tax hits wealth, not income. Homeowners don't have extra cash to pay the tax. They must cut consumption (MPC out of tax = 0.27 for cars). People show strong debt aversion — they won't borrow against their home equity even though they could. Bottom line: property tax is effectively America's wealth tax.

6. Racial Disparities in Property Assessment (Avenancio-León & Howard 2022)

Black and Hispanic homeowners face systematically higher property tax assessments relative to actual home value. Median Black homeowner pays \$350 extra/year. Two mechanisms: (1) coarse assessment

formulas, (2) whites more likely to successfully appeal assessments. Solution: using automated valuation (Zillow) eliminates the racial gap.

7. Intergovernmental Grants

Grant Type	How It Works	Effect
Matching grant	Federal matches local spending dollar-for-dollar	Changes relative price of public good. Both substitution + income effects. Strongest effect.
Block grant	Fixed amount, spend however you want	Only income effect (parallel budget shift). Weaker effect.
Conditional block	Fixed amount, must spend on X	Same as block grant unless binding constraint.

Flypaper effect: "Money sticks where it lands" — grants don't crowd out local spending as much as theory predicts. Short-run effect, may fade long-run.

True / False / Uncertain — Master Practice Table

These are high-probability T/F/U statements based on lecture content. For EVERY answer, give the verdict AND a 2-sentence explanation of the mechanism.

Statement	Verdict	Key Reasoning
A dividend tax cut should increase corporate investment.	FALSE	New view: dividend tax affects old/new investment symmetrically for mature firms. Yagan (2015) found zero investment response to 2003 cut.
If the government taxes business profits net of all costs, investment is undistorted.	TRUE	Net profit tax: FOC is $F'(K)=r$, same as no-tax case. Tax only takes share of pure profit without changing investment incentive.
A capital income tax always reduces savings.	FALSE	Substitution effect reduces savings but income effect increases savings. Net effect is theoretically ambiguous.
A consumption tax is equivalent to a labor income tax.	TRUE	Both reduce budget uniformly across time. Neither distorts savings decision. Only capital income tax specifically penalizes future consumption.
The Coase theorem implies government intervention is never needed for externalities.	FALSE	Coase requires well-defined property rights, costless bargaining, few parties. Fails for large-scale externalities (climate change).
Taxes are better than permits for addressing climate change.	TRUE (mostly)	Climate change has relatively flat MD curve (each ton CO_2 does similar damage). Taxes lock in price efficiently. Permits better for steep MD.
Free markets underprovide public goods.	TRUE	Non-excludability $\rightarrow$ free riding. Each person ignores benefit to others. Private provision < socially optimal level.
Government grants to local communities fully crowd out private spending.	FALSE	Flypaper effect: money sticks where it lands. Empirically, crowd-out is partial, not complete.
The Tiebout model shows local government can provide public goods efficiently.	TRUE (with caveats)	Voting with feet creates competition. But requires: perfect mobility, information, no spillovers, poll taxes. Reality doesn't fully satisfy these.
Property taxes are popular because they fund local schools.	FALSE	Property tax is America's most hated tax (Wong 2024). Hits wealth not income, creates cash flow problems, people hate borrowing.
The EITC is most effective when extensive margin responses dominate.	TRUE	EITC subsidizes entering workforce (extensive). Its phase-out hurts intensive margin. Best when participation response is large.
Most taxable income response from top earners is real labor supply change.	FALSE	Much is avoidance: income shifting, timing of capital gains, entity classification. Goolsbee (2000) showed executive option timing.
Buildings tend to have too few murals painted on their sides (from efficiency perspective).	TRUE	Murals are positive consumption externality (passersby enjoy them, owner uncompensated). Positive externalities $\rightarrow$ underprovision.
Property tax assessments are race-neutral in the US.	FALSE	Avenancio-León & Howard (2022): Black/Hispanic homeowners systematically

	over-assessed. Median Black homeowner pays \$350 extra/year.
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Complete Formula Reference Sheet

Per syllabus: you don't need to memorize formulas derived in class, but you DO need to know the concepts they communicate. For the analytic section, you must be able to set up and solve problems from scratch.

Concept	Formula	Key Note
Budget constraint (HH)	$c = (1-t)wl + T$	After-tax wage $\times$ labor + transfer
Optimal labor supply	$l^* = [(1-t)w]^\eta$	FOC of standard utility model
Optimal consumption	$c^* = [(1-t)w]^\eta(1+\eta) + T$	Plug l^* into budget
Taxable income elasticity	$TIE = \eta$	Empirically 0.1–0.5
Revenue-max rate (Laffer)	$\tau^* = 1/(1+e)$	Higher $e \rightarrow$ lower optimal rate
Optimal top rate	$\tau^* = 1/(1+ae)$	$a \approx 1.5$, $e \approx 0.25\text{--}0.5$
Optimal linear tax	$\tau = (1-\bar{a})/(1-\bar{a}+e)$	Close to Laffer rate when inequality high
Firm FOC (no tax)	$F'(K) = r$	Marginal product = cost of capital
Gross profit tax	$F'(K) = r/(1-\tau)$	Reduces investment
Net profit tax	$F'(K) = r$	No distortion (full deduction)
Traditional view (dividend)	$F'(K) = r/[(1-\tau_D)(1-\tau_C)]$	Dividend tax hurts investment
New view (dividend)	$F'(K) = r/(1-\tau_C)$	Dividend tax irrelevant for investment
IBC (no tax)	$c_1 + c_2/(1+r) = w \cdot l$	Intertemporal budget constraint
IBC (capital tax)	$c_1 + c_2/[1+r(1-\tau_K)] = w \cdot l$	Only r is affected by τ_K
Wealth accumulation	$W_t = (1+r)W_{t-1} + E_t - C_t + I_t$	Returns + earnings – consumption + inheritance
Wealth/income ratio	$\beta = s/g$	s = savings rate, g = growth rate
Social marginal cost	$SMC = PMC + MD$	Add external damage to private cost
DWL from externality	$\frac{1}{2} \times MD \times (Q_{\text{mkt}} - Q_{\text{opt}})$	Triangle area
Corrective tax	$t^* = MD$	Set tax = marginal damage
Samuelson rule	$\sum MRS_i = MC$	Sum of individual valuations = cost
DD regression	$Y = \alpha + \beta T + \gamma \text{Post} + \delta(T \times \text{Post})$	δ = causal DD estimate
Debt dynamics	$B(t+1) = (1+r)B(t) + PD(t)$	$PD = G - T$ (primary deficit)
Tax revenue (Laffer)	$TR(\tau) = \tau(1-\tau)^\eta \varphi$	Taxable income = $(1-\tau)^\eta \varphi$
Consumer incidence share	$\varepsilon_s / (\varepsilon_s - \varepsilon_d)$	$\varepsilon_s > 0$, $\varepsilon_d < 0$; less elastic bears more
DWL (tax incidence)	$\frac{1}{2} \times t \times (Q_0 - Q_1)$	Triangle between S and D, or $\propto t^2 \times \text{elasticity}$

Key Empirical Studies Reference

You should know the study, the method, and the main finding for each. These appear in T/F/U and empirical sections.

Study	Question	Method	Key Finding
Yagan (2015)	Does dividend tax cut boost investment?	DD: C-corps vs. S-corps around 2003 cut	Elasticity = 0.00. New view wins.
Chodorow-Reich et al. (2025)	2017 tax cut investment effect?	DD: firms with larger cost-of-capital reduction	Treatment firms increased investment more.
Chay-Greenstone (2003)	Health effects of Clean Air Act?	DD: counties above vs. below pollution threshold	~500 fewer infant deaths per 100K.
Barreca et al. (2016)	Heat wave mortality over time?	Temperature-mortality regression, 1900–2004	75% decline in heat mortality. AC adoption.
KLS (2013)	Tax effects on migration?	DD: Danish tax break for high-earning foreigners	Sharp increase in foreign workers. Large mobility response.
Saez (2010)	Bunching at EITC kinks?	Income distribution analysis	Bunching only among self-employed. Wage workers can't manipulate.
Goolsbee (2000)	Executive response to tax hike?	Timing of stock option exercises around 1993 reform	Large retiming, small real response.
Hungerman (2005)	Church welfare crowding out?	DD: 1996 welfare reform, high vs. low non-citizen areas	40 cents church increase per \$1 welfare cut.
Andreoni-Payne (2003)	Government grants crowd out private giving?	Panel data on nonprofits	\$1 grant = 56c less private. 70% from reduced fundraising.
Wong (2024)	Why do people hate property tax?	Property tax reassessments	MPC = 0.27 on cars. Cash flow problem, debt aversion.
Avenancio-León & Howard (2022)	Racial bias in property assessment?	Assessment-to-sale ratio by race	Black homeowners over-assessed. \$350/yr extra.
Cellini et al. (2010)	School bonds affect house values?	RD: bonds that just passed vs. just failed	House prices up 3–9%. Schools were under-invested.

Exam Strategy & Tips

General Tips

LESSON FROM YOUR MIDTERM 1 PERFORMANCE

Your main issue was algebra execution under time pressure (cascading errors in Q3 cost you ~30 points). For Midterm 2: (1) Write each step slowly and clearly. (2) Double-check arithmetic before moving to the next part. (3) If you get a weird number, re-check — don't keep going with a bad answer. (4) Partial credit is generous, so ALWAYS show your setup even if you can't finish.

T/F/U Section (60 pts — ~20 min)

Each statement is worth 10 points. Give the verdict (True/False/Uncertain) AND a 2-sentence explanation. The explanation matters more than the verdict. Structure: **Verdict.** Reason why. Key mechanism or formula.

Common traps: "Always" and "never" statements are usually FALSE. "Uncertain" is valid when a result depends on parameter values (like income vs. substitution effects on savings).

Empirical Section (30 pts — ~15 min)

You will likely be given a policy scenario and asked to set up the DD regression. Steps:

1. Identify treatment and control groups
2. Write: $Y_{it} = \alpha + \beta \cdot \text{Treated}_i + \gamma \cdot \text{Post}_t + \delta \cdot (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$
3. Interpret δ as the causal DD estimate
4. Discuss parallel pre-trends assumption and any threats
5. If given a graph, read off approximate values for the 2×2 table and compute δ

Analytic Section (60 pts — ~35 min)

Likely topics: externality problem (SMC/PMC, tax/permit), business tax (FOC with/without deduction), or savings model (two-period, capital tax effects). Steps:

1. Write the objective function (what is being maximized)
2. Write the constraint (budget constraint)
3. Substitute or use Lagrangian
4. Take FOC, solve step by step
5. Label everything: state what each variable represents

Time Allocation

Section	Time	Strategy
T/F/U (60 pts)	~20 min	~3 min per statement. Verdict + 2 sentences. Move fast. Don't overthink.
Empirical (30 pts)	~15 min	Write regression, interpret coefficients, discuss assumptions. Draw the 2×2 table.
Analytic (60 pts)	~35 min	Show ALL steps. Set up, FOC, solve. Check arithmetic. Partial credit is generous.
Buffer	~5 min	Review answers. Check for silly arithmetic errors.

Good luck on the exam, Marcus! — Generated by Claude